

1. Java Basics

- What is Java?
- Features of Java
- History of Java
- Java Editions (SE, EE, ME)
- JVM, JRE, JDK (architecture & flow)
- Java program structure
- `main()` method
- Compilation & execution process
- Java keywords & identifiers
- Comments in Java

2. Data Types & Variables

- Primitive data types
(`byte`, `short`, `int`, `long`, `float`, `double`, `char`, `boolean`)
- Wrapper classes
- Type casting (implicit & explicit)
- Variables
 - Local
 - Instance
 - Static
- Literals
- Constants (`final`)

3. Operators

- Arithmetic operators
- Relational operators
- Logical operators
- Bitwise operators

- Assignment operators
- Unary operators
- Ternary operator
- Operator precedence & associativity

4. Control Statements

Conditional Statements

- `if`
- `if-else`
- `else-if` ladder
- Nested `if`
- `switch`

Looping Statements

- `for`
- `while`
- `do-while`
- `for-each`

Jump Statements

- `break`
- `continue`
- `return`

5. Arrays

- One-dimensional arrays
- Two-dimensional arrays
- Multidimensional arrays
- Array initialization

- Array length
- Array vs Collection
- `Arrays` utility class

◆ 6. Methods

- Method definition
- Method invocation
- Return type
- Method parameters
- Call by value
- Static vs instance methods
- Method overloading
- Recursion
- Var-args methods

◆ 7. Object-Oriented Programming (OOP)

- Class & Object
- Constructors
 - Default constructor
 - Parameterized constructor
 - Constructor overloading
- `this` keyword
- `static` keyword
- `final` keyword
- `super` keyword

◆ 8. OOP Concepts in Detail

Inheritance

- Types of inheritance
- `extends` keyword
- Method overriding
- IS-A relationship

Polymorphism

- Compile-time polymorphism
- Runtime polymorphism
- Method overriding rules

Abstraction

- Abstract classes
- Abstract methods
- Interfaces
- Difference between abstract class & interface

Encapsulation

- Data hiding
- Getters & setters
- Access control

◆ 9. Access Modifiers

- `public`
- `protected`
- `default`
- `private`
- Access scope table

◆ 10. Relationships in Java

- Association
- Aggregation
- Composition
- Inheritance vs Composition (interview favorite)

◆ 11. Strings

- `String` class
- String immutability
- `StringBuffer`
- `StringBuilder`
- String methods
- String comparison
- String memory (String pool)

◆ 12. Wrapper Classes & Autoboxing

- Primitive vs Wrapper
- Autoboxing
- Unboxing
- Use cases

◆ 13. Exception Handling

- What is an exception?
- Types of exceptions
 - Checked
 - Unchecked

- Errors
- `try`
- `catch`
- `finally`
- `throw`
- `throws`
- Custom exceptions
- Exception hierarchy

◆ 14. File Handling (`java.io` & `java.nio`)

- File class
- Reading files
- Writing files
- Byte streams
- Character streams
- Serialization & Deserialization

◆ 15. Multithreading

- Process vs Thread
- Thread life cycle
- Creating threads
 - `Thread` class
 - `Runnable` interface
- Synchronization
- Inter-thread communication
- `sleep()`, `join()`, `wait()`, `notify()`
- Thread safety

◆ 16. Collections Framework

- List
 - ArrayList
 - LinkedList
 - Vector
- Set
 - HashSet
 - LinkedHashSet
 - TreeSet
- Map
 - HashMap
 - LinkedHashMap
 - TreeMap
 - Hashtable
- Iterator & ListIterator
- Comparable & Comparator
- Collection vs Collections

◆ 17. Generics

- Generic classes
- Generic methods
- Bounded types
- Wildcards
- Type safety

◆ 18. Java 8 Features (Core Java Level)

- Lambda expressions
- Functional interfaces
- Stream API
- Method references
- Default methods
- Static methods in interface
- Optional class

◆ 19. Inner Classes

- Static inner class
- Non-static inner class
- Local inner class
- Anonymous inner class

◆ 20. Object Class Methods

- `toString()`
- `equals()`
- `hashCode()`
- `clone()`
- `finalize()`
- `getClass()`

◆ 21. JVM Internals (Core Level)

- Class loader
- Memory areas
- Garbage collection
- Stack vs Heap
- JVM architecture

◆ 22. Important Interview Topics

- `==` vs `equals()`
- `HashMap` internal working
- Immutable class
- Singleton class
- Marker interfaces
- Serialization vs Deserialization
- Deep copy vs Shallow copy